Homework

How bad can things get? We consider daily closing prices for the S&P 500 index from 11/1/2011 until 12/29/2021. For our purposes, the total number of days is $N + 1 = 2556$ which generates $N = 2555$ log returns. If not, please adjust your end date to match the total number. In this exercise we will see what our extreme value theory-based risk measures have to say about the potential for a crash.

Consider a hypothetical portfolio where each day we start with \$1,000,000 in the S&P 500 index. The daily losses are estimated using linearized operator so that $L^{\text{lin}} = -VX$ where V is the portfolio value, and X is the daily log return. For each of the methods described below, determine the VaR_α, as a function of α, for the portfolio losses.

For each method, produce a plot of $\alpha \mapsto \text{VaR}_\alpha$ for α between .99 and .9999 in increments of .000099 (100 values). For $\alpha = .9999$ what are the VaR_α values? Which is highest? How do the two EVT estimates compare (is there a meaningful difference)?

(1) Using the empirical distribution for the log returns.

(2) Assuming that as of $t = 11/1/2021$ the log returns $X_{t+\Delta}$ over the next business day are normally distributed with mean $\mu_{t+\Delta}$ and variance $\sigma^2_{t+\Delta}$. To estimate $\mu_{t+\Delta}, \sigma^2_{t+\Delta}$ use an EWMA procedure with parameter $\lambda = \theta = 0.97$ and a 100 day initialization time.

(3) Assuming a GEV distribution for the maximum and the block maximum method. For the $N+1 = 2556$ days of data (which generate $N = 2555$ log returns), break the data into blocks of size (i) $n = 73$ days for a total of $m = 35$ blocks and (ii) $n = 35$ days for a total of $m = 73$ blocks. Estimate the resultant GEV parameters using the function provided in scipy.

(4) Assume a GP excess loss distribution. Here, take $u = \text{VaR}_{.95}$ as estimated in (1) above. To estimate the GP parameters, you will have to select only those losses above the u threshold, and compute the loss minus u for these selected days. With this data you can then fit the GP distribution using scipy provided functions. With the fitted parameters, estimate the VaR_α in terms of α and $F(u)$ which in this case is 0.95 by construction.